Homework Project 2

1 State Estimation and Source Seeking with Particle Filter

Problem 1 Consider a large open two-dimensional space in which a robot can freely move. In this space, there is a single moving light source, which the robot aims to follow and stay as close as possible to it. This defines a so-called source seeking problem. In Figure 1 we can observe an example case for a source seeking problem, where the source (the orange dot) moves in the two-dimensional space (x_1, x_2), and is tracked by the robot (the blue dot).

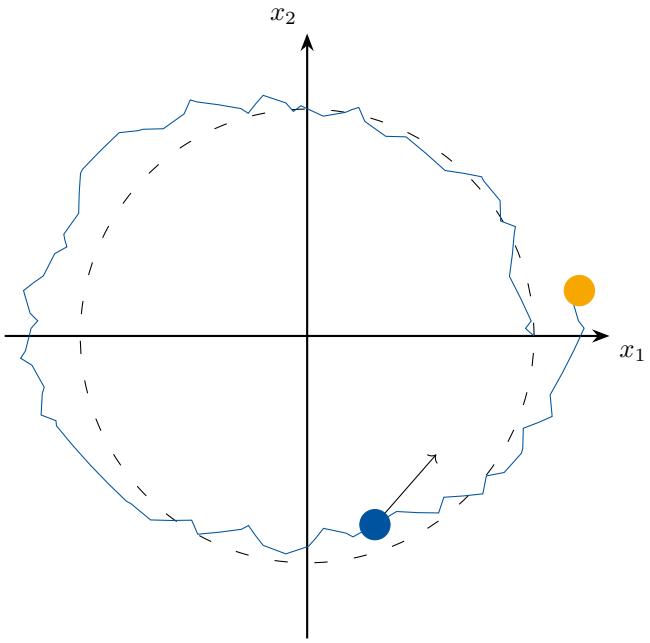


Figure 1: A robot (blue dot) tracking the light source (orange dot) in the (x_1, x_2) space

In the following we model the dynamics of the robot and the motion of the light source (lamp). We assume that the robot position can be measured exactly at any time and its position is controlled by two inputs $u \in \mathbb{R}^2$. The robot is fully actuated, however the control inputs are constrained to ensure the robot has a finite velocity, and thereby its position θ is given by

$$\dot{\theta}(t) = u(t), \qquad \|u(t)\|_2^2 \leq 1. \tag{1.1}$$

The position $\xi = [\xi_1 \ \xi_2]^\top$ of the lamp is governed by the stochastic differential equation (SDE)

$$\begin{bmatrix} \mathrm{d}\xi_1(t) \\ \mathrm{d}\xi_2(t) \end{bmatrix} = 0.2 \begin{bmatrix} -\sin\left(\frac{t}{50}\right) \\ +\cos\left(\frac{t}{50}\right) \end{bmatrix} \mathrm{d}t + \begin{bmatrix} 1 \\ 1 \end{bmatrix} \mathrm{d}w_L(t), \tag{1.2}$$

where $\mathrm{d}w_L$ is a Laplace distributed process. To avoid the difficulties of dealing with an SDE, we analyze a discrete-time model, i.e. we set $t_k := k\delta t$ for some small $\delta t > 0$, and thus we have

$$\frac{1}{\delta t} \begin{bmatrix} \xi_1(t_{k+1}) - \xi_1(t_k) \\ \xi_2(t_{k+1}) - \xi_2(t_k) \end{bmatrix} = 0.2 \begin{bmatrix} -\sin\left(\frac{t_k}{50}\right) \\ +\cos\left(\frac{t_k}{50}\right) \end{bmatrix} + \frac{1}{\delta t} \begin{bmatrix} w_1(t_k) \\ w_2(t_k) \end{bmatrix}, \tag{1.3}$$

where $w_1(t_k), w_2(t_k) \sim \text{Laplace}(0, \sigma)$ are i.i.d. disturbances. These stochastic disturbances represent a way to model the non-predictable behavior of the lamp. The robot obtains noisy measurements of the lamp position, which are modeled by the equations

$$\begin{bmatrix} \zeta_1(t_k) \\ \zeta_2(t_k) \end{bmatrix} = \begin{bmatrix} \xi_1(t_k) + v_1(t_k) \\ \xi_2(t_k) + v_2(t_k) \end{bmatrix},$$

where $v_1(t_k), v_2(t_k) \sim \mathcal{U}[-\varepsilon, \varepsilon]$, with $\varepsilon > 0$, is the measurement noise at every time instance $t_k \geq 0$, which is i.i.d. By the introduction of the measurement noise we represent the uncertainty the robot has about the position of the lamp.

We approach the source seeking problem in two steps. In the first step, we address the problem of estimating the position of the lamp at every time step $t_k = k\delta t$, $k \in \mathbb{N}_0$, $\delta t > 0$, through the use of a Particle Filter (PF). In the second step, we address the problem of steering the robot to the current estimate of the lamp position (the estimate obtained by using the PF).

The first task is to define the state equations and output equations for the position estimation problem of the lamp. Here, we consider the discrete-time setting, i.e. $t_k = k\delta t$. In particular, provide:

1a) the state equations for the lamp dynamics and the corresponding density $\rho(\xi(t_{k+1})|\xi(t_k), t_k)$ in the form of

$$\xi(t_{k+1}) = f_1(\xi(t_k), t_k) + g_1(w(t_k)), \qquad \rho(\xi(t_{k+1})|\xi(t_k), t_k) = \rho(f_1(\xi(t_k), t_k)|\xi(t_k), t_k),$$

1b) the output equations (measurement equations) of lamp position and the corresponding density $\rho(\zeta(t_k)|\xi(t_k))$ in the form of

$$\zeta(t_k) = h_1(\xi(t_k)) + d_1(v(t_k)), \qquad \rho(\zeta(t_k)|\xi(t_k)) = \rho(h_1(\xi(t_k))|\xi(t_k)).$$

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Problem 2 In this problem, we will have a look at the derivation and analysis of the Particle Filter (PF). The PF, like the Kalman filter (KF), is based on the Bayes' filter and aims to determine the posterior distribution of the states at every time instance (i.e. time step), given the inputs and outputs. In general, the KF may only work with linear systems, while the Bayes' filter does not have such restrictions, and may work with any arbitrary distribution and nonlinear dynamics. However, the Bayes filter is in general computationally intractable. The PF can be seen as a computationally tractable approximation of the Bayes filter by approximating the states' posterior distribution with a finitely supported distribution.

For the derivation of the PF, we consider discrete-time state space models with $x_k \in \mathbb{R}^d$ representing the state at time k, $u_k \in \mathbb{R}^q$ the input, and $z_k \in \mathbb{R}^p$ representing the output:

$$\begin{aligned} x_k &= f(x_{k-1}, u_{k-1}) + \nu_{k-1} \\ z_k &= g(x_k) + v_k. \end{aligned}$$

Here, ν_k and v_k represent zero-mean disturbances with finite variance. In addition, for the sake of simplicity, we assume that g is independent of u_k. For this system description, we aim to use the PF to obtain an estimate in terms of the conditional probability density function $\hat{\rho}(x_k|u_{0:k-1}, z_{1:k})$ of the state at every $k \geq 1$ given measurements z_1 to z_k and a prior density $\rho(x_0)$ of the initial state. The inputs are assumed to be deterministic and known. Here, we denote with $u_{0:k-1}$ and $z_{1:k}$ the arrays (sets) of inputs and measurements from time instance $i = 0$ to $i = k-1$ and $i = 1$ to $i = k$, respectively.

Analogous to the Bayes' filter, the PF has two core steps, namely the predict and the update step (and in addition it has a resampling step, which is discussed below). The prediction step is used to form a prediction of the states' distribution for the next time step given the past information (past measurements), and the update step updates the predicted distribution by utilizing the current (new) measurement. In more detail, like the Bayes' filter, the PF is obtained by a recursive application of Bayes's rule, i.e.

$$\rho(x_k|u_{0:k-1}, z_{1:k}) = \frac{\rho(z_k|x_k)}{\rho(z_k|u_{0:k-1}, z_{1:k-1})} \rho(x_k|u_{0:k-1}, z_{1:k-1}),$$

where in red we have the prediction term, and in blue we have the update term. The core idea of the PF is to approximate both the prediction and update terms, which is particularly important in high-dimensional problems where an explicit solution of the Bayes' filter is hard to obtain (at least in real-time). This is done by sampling, i.e., the conditional probability density function

of the state x_k at time k given the inputs $u_{0:k-1}$ and measurements $z_{1:k}$ is approximated by a measure supported on a finite set:

$$\rho(x_k|u_{0:k-1}, z_{1:k}) \approx \sum_{i=1}^{N} w_k^{(i)}(u_{0:k-1}, z_{1:k})\delta(x_k - x_k^{(i)}) =: \hat{\rho}(x_k|u_{0:k-1}, z_{1:k}), \tag{2.1}$$

where $w_k^{(i)} := w_k^{(i)}(u_{0:k-1}, z_{1:k}) > 0$ is the so-called i-th (importance) weight at time k, $x_k^{(i)}$ is the so-called i-th particle (i-th sample) at time step k, and $\delta(\cdot)$ is the Dirac function (Dirac measure). Thus $\hat{\rho}(x_k|u_{0:k-1}, z_{1:k})$ defines an approximation of the density $\rho(x_k|u_{0:k-1}, z_{1:k})$ by using finitely many particles and weights, as illustrated in Figure 2.

With a slight abuse of notation, we denote in the following the time-dependent (conditional) probability density (mass) functions as $\rho(x_k|z_{1:k}) := \rho_{x_k}(\cdot|z_{1:k})$. Similarly, we use the notation of $\rho(x_k|u_{0:k-1}, z_{1:k-1}) := \rho_{x_k}(\cdot|u_{0:k-1}, z_{1:k-1})$ and $\rho(z_k|x_k) := \rho_{z_k}(\cdot|x_k)$, and so on. We moreover denote by $x \sim \mathcal{D}\{x_1, ..., x_n\}$ a sample drawn from the distribution $\mathcal{D}$ with finite support $x_1, ..., x_n$ and probability mass function $p(x = x_i) = p_i$, $\sum_{i=1}^{n} p_i = 1$. Similarly, $x \sim \rho(\cdot)$ denotes a sample drawn from probability density function $\rho(\cdot)$ with $\int_{-\infty}^{\infty} \rho(x)dx = 1$.

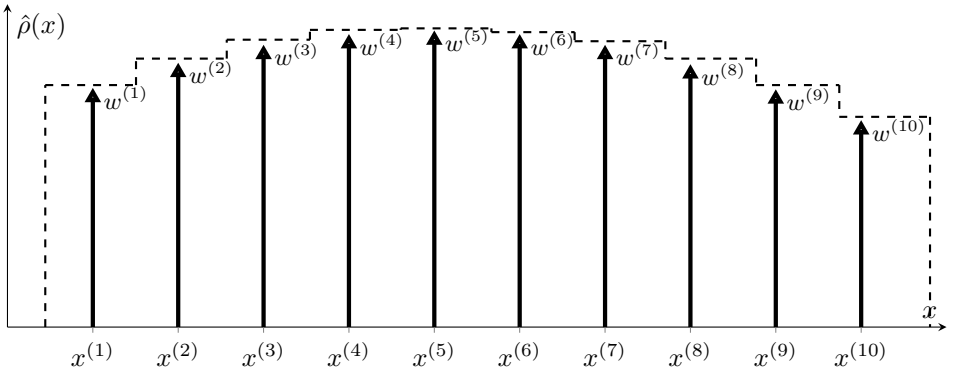


Figure 2: Illustration of finitely supported Dirac measures $\hat{\rho}(x) = \sum_{i=1}^{10} w^{(i)}\delta(x - x^{(i)})$. $x^{(i)}$ is called the i-th particle and $w^{(i)}$ the corresponding particle weight. The dashed lines visualize a possible density (histogram representation) approximated by the finitely supported Dirac measures.

The prediction step and update step rely on so-called sequential importance sampling methods. Thus, we start the derivation of the PF by firstly summarizing importance sampling methods and Monte-Carlo methods.

Monte-Carlo integration with importance sampling. Let ξ be a continuous real-valued random variable with a probability density function (in short, density) ρ_ξ. The goal is to approximately compute the expected value (integral) $\mathbf{E}(\xi)$ when it is too difficult to draw samples from ρ_ξ. Suppose a density $\mu(\cdot)$ is known which is assumed to be a sufficiently good approximation of $\rho_\xi(\cdot)$ and from which samples can be drawn efficiently. Let $\xi^{(i)}$ $i \in \{1, \dots, N\}$ be samples drawn according to the density $\mu(\cdot)$, which is called the importance density, and for which it is assumed that it is non-zero wherever $\rho_\xi(\cdot)$ is non-zero. Then the idea is

$$\mathbf{E}(\xi) = \int_{-\infty}^{\infty} \tau\rho_\xi(\tau)\mathrm{d}\tau = \int_{-\infty}^{\infty} \tau\rho_\xi(\tau)\frac{\mu(\tau)}{\mu(\tau)}\mathrm{d}\tau = \int_{-\infty}^{\infty} \tau\frac{\rho_\xi(\tau)}{\mu(\tau)}\mu(\tau)\mathrm{d}\tau \approx \sum_{i=1}^{N} \xi^{(i)}\frac{\rho_\xi(\xi^{(i)})}{\mu(\xi^{(i)})} =: \hat{\xi}.$$

Moreover, we can represent this approximation by

$$\hat{\xi} = \int_{-\infty}^{\infty} \tau \sum_{i=1}^{N} \frac{\rho_\xi(\tau)}{\mu(\tau)}\delta\left(\tau - \xi^{(i)}\right)\mathrm{d}\tau =: \int_{-\infty}^{\infty} \tau\hat{\rho}_\xi(\tau)\mathrm{d}\tau.$$

The terms $\frac{\rho_\xi(\xi^{(i)})}{\mu(\xi^{(i)})}$ evaluated at every sample $\xi^{(i)}$ represent corresponding weights. Furthermore, we observe that the approximation of the density of ξ is given by $\hat{\rho}_\xi(\tau) := \sum_{i=1}^{N} \frac{\rho_\xi(\tau)}{\mu(\tau)}\delta\left(\tau - \xi^{(i)}\right)$.

Notice, if ρ_ξ was easy to sample from, we would set $\mu = \rho_\xi$, thus every weight would be one. This special case corresponds to Monte-Carlo integration. In general, this procedure is called Monte-Carlo integration with importance sampling and is used in the PF.

Initialization step. We will now start with the initialization of the PF. Typically, the prior density $\rho(\cdot)$ of the initial state x_0 is known, and thus N samples, which are called particles, are sampled as follows

$$x_0^{(i)} \sim \rho(\cdot), \quad i = 1, \dots, N.$$

In addition, all the weights are initialized to be identical, i.e.

$$w_0^{(i)} = \frac{1}{N}, \quad i = 1, \dots, N.$$

Prediction step. We will now have a look at the prediction step of the PF. As in the Bayes' filter, the idea here is to make a prediction for the density of the state x_k at time k given measurements up to time $k-1$, i.e. we aim to compute $\rho(x_k|u_{0:k-1}, z_{1:k-1})$. Using the Chapman-Kolmogorov equation, we have

$$\rho(x_k|u_{0:k-1}, z_{1:k-1}) = \int_{x_{k-1}} \rho(x_k|x_{k-1}, u_{k-1})\rho(x_{k-1}|u_{0:k-1}, z_{1:k-1})\mathrm{d}x_{k-1}.$$

In the first step, we approximate the left hand side by

$$\rho(x_k|u_{0:k-1}, z_{1:k-1}) \approx \sum_{i=1}^{N} \tilde{w}_k^{(i)} \delta(x_k - x_k^{(i)}) =: \tilde{\rho}(x_k|u_{0:k-1}, z_{1:k-1}),$$

where x_k and x_{k-1} are the states at time steps k and $k-1$ respectively, $u_{0:k-1}$ is the set of inputs from time step 0 to $k-1$ and $x_k^{(i)}$ and $\tilde{w}_k^{(i)}$ are the i-th sample and weight, respectively. Here, the weights have to be finite and non-negative. Next, in order to compute the weights and suitable samples, we introduce the importance density $\mathrm{q}(\cdot)$. The importance density is typically defined as a density, which is non-zero where $\rho(\cdot)$ is non-zero and allows for easy sampling (in some cases it is not possible to sample over the whole domain). In more detail, we want to approximate the density by using finitely many samples, i.e.

$$\begin{aligned}\rho(x_k|u_{0:k-1}, z_{1:k-1}) &= \frac{\rho(x_k|u_{0:k-1}, z_{1:k-1})}{\mathrm{q}(x_k|x_{k-1}, u_{k-1})}\mathrm{q}(x_k|x_{k-1}, u_{k-1}) \\ &\approx \sum_{i=1}^{N} \frac{\rho(x_k^{(i)}|u_{0:k-1}, z_{1:k-1})}{\mathrm{q}(x_k^{(i)}|x_{k-1}, u_{k-1})}\delta(x_k - x_k^{(i)}),\end{aligned}$$

with $x_k^{(i)}$ sampled from $\mathrm{q}(\cdot|x_{k-1}, u_{k-1})$. Observe that the approximation in the second line corresponds to the density estimate in the Monte Carlo integration section given above. We define the corresponding weights above as

$$\tilde{w}_k^{(i)} := \frac{\rho(x_k^{(i)}|u_{0:k-1}, z_{1:k-1})}{\mathrm{q}(x_k^{(i)}|x_{k-1}^{(i)}, u_{k-1})}, \quad i = 1, \dots, N,$$

where the particles $x_k^{(i)}$ are predictions of the next state using the importance density, i.e. we sample N particles as follows

$$x_k^{(i)} \sim \mathrm{q}(\cdot|x_{k-1}^{(i)}, u_{k-1}), \quad i = 1, \dots, N,$$

Consequently, we obtain an approximate density of the form

$$\tilde{\rho}(x_k|u_{0:k-1}, z_{1:k-1}) = \sum_{i=1}^{N} \tilde{w}_k^{(i)} \delta(x_k - x_k^{(i)}),$$

which is, again, motivated by the Monte Carlo integration with importance sampling method.

It remains to determine (approximately) the weights $\tilde{w}_k^{(i)}$. Notice that $\rho(x_k^{(i)}|u_{0:k-1}, z_{1:k-1})$ is not known, since it is the density we aim to determine. Therefore we cannot evaluate the above quotients directly and we need to go a step further and turn now to the right hand side of the Chapman-Kolmogorov equation. To this end, the term $\rho(x_{k-1}|u_{0:k-1}, z_{1:k-1})$ in the integrant of the Chapman-Kolmogorov equation

$$\rho(x_k^{(i)}|u_{0:k-1}, z_{1:k-1}) = \int_{x_{k-1}} \rho(x_k^{(i)}|x_{k-1}, u_{k-1})\rho(x_{k-1}|u_{0:k-1}, z_{1:k-1})\mathrm{d}x_{k-1},$$

will be approximated as follows

$$\rho(x_{k-1}|u_{0:k-1}, z_{1:k-1}) = \rho(x_{k-1}|u_{0:k-2}, z_{1:k-1}) \approx \sum_{i=1}^{N} w_{k-1}^{(i)}\delta(x_{k-1} - x_{k-1}^{(i)}).$$

Consequently, using this approximation and plugging the right hand side of the Chapman-Kolmogorov into the quotient defining $\tilde{w}_k^{(i)}$, we get after integration over x_{k-1} (using the Dirac choosing property), the following expression for the weights:

$$\begin{aligned}\tilde{w}_k^{(i)} \approx \hat{\tilde{w}}_k^{(i)} :&= \frac{\int_{x_{k-1}} \rho(x_k^{(i)}|x_{k-1}, u_{k-1}) \sum_{i=1}^{N} w_{k-1}^{(i)}\delta(x_{k-1} - x_{k-1}^{(i)})\mathrm{d}x_{k-1}}{\mathrm{q}(x_k^{(i)}|x_{k-1}^{(i)}, u_{k-1})} \\ &= \frac{\rho(x_k^{(i)}|x_{k-1}^{(i)}, u_{k-1})w_{k-1}^{(i)}}{\mathrm{q}(x_k^{(i)}|x_{k-1}^{(i)}, u_{k-1})}, \quad i = 1, \ldots, N.\end{aligned}$$

Summarizing, the approximated predicted posterior density $\hat{\rho}(x_k|u_{0:k-1}, z_{1:k-1})$ is thus given as

$$\hat{\rho}(x_k|u_{0:k-1}, z_{1:k-1}) = \sum_{i=1}^{N} \hat{\tilde{w}}_k^{(i)}\delta(x_k - x_k^{(i)}).$$

Update step. In the following, we present the (derivation of the) update step of the PF, which aims to obtain the estimate $\hat{\rho}(x_k|u_{0:k-1}, z_{1:k})$, i.e. to update the density estimate of the prediction step w.r.t. the new information (new measurement) z_k. The key idea of this step is to use the Bayes' rule for conditional probabilities, and to incorporate the prediction step from above.

Remember that the considered system is a Markov system, and the output z_k depends solely on x_k. Hence, from the Bayes' rule we have

$$\rho(x_k|u_{0:k-1}, z_{1:k}) = \frac{\rho(z_k|x_k)\rho(x_k|u_{0:k-1}, z_{1:k-1})}{\int_{x_k} \rho(z_k|x_k)\rho(x_k|u_{0:k-1}, z_{1:k-1})\mathrm{d}x_k},$$

where we plug in $\hat{\rho}(x_k|u_{0:k-1}, z_{1:k-1})$ instead of $\rho(x_k|u_{0:k-1}, z_{1:k-1})$, and by using the above given predicted posterior density relation we obtain

$$\hat{\rho}(x_k|u_{0:k-1}, z_{1:k}) = \frac{\rho(z_k|x_k) \sum_{i=1}^{N} \hat{\tilde{w}}_k^{(i)}\delta(x_k - x_k^{(i)})}{\int_{x_k} \rho(z_k|x_k) \sum_{j=1}^{N} \hat{\tilde{w}}_k^{(j)}\delta(x_k - x_k^{(j)})\mathrm{d}x_k}.$$

By the choosing property of the Dirac function, we get

$$\hat{\rho}(x_k|u_{0:k-1}, z_{1:k}) = \sum_{i=1}^{N} \frac{\rho(z_k|x_k^{(i)})\hat{\tilde{w}}_k^{(i)}\delta(x_k - x_k^{(i)})}{\sum_{j=1}^{N} \rho(z_k|x_k^{(j)})\hat{\tilde{w}}_k^{(j)}} =: \sum_{i=1}^{N} w_k^{(i)}\delta(x_k - x_k^{(i)}),$$

where from the relation for $\hat{\tilde{w}}_k^{(i)}$ we get

$$w_k^{(i)} := \frac{\rho(z_k|x_k^{(i)})\hat{\tilde{w}}_k^{(i)}}{\sum_{j=1}^{N}\rho(z_k|x_k^{(j)})\hat{\tilde{w}}_k^{(j)}} = \frac{\rho(z_k|x_k^{(i)})\frac{\rho(x_k^{(i)}|x_{k-1}^{(i)},u_{k-1})}{\mathrm{q}(x_k^{(i)}|x_{k-1}^{(i)},u_{k-1})}}{\sum_{j=1}^{N}\rho(z_k|x_k^{(j)})\frac{\rho(x_k^{(j)}|x_{k-1}^{(j)},u_{k-1})w_{k-1}^{(j)}}{\mathrm{q}(x_k^{(j)}|x_{k-1}^{(j)},u_{k-1})}}w_{k-1}^{(i)}$$
$$= \frac{\rho(z_k|x_k^{(i)})\frac{\rho(x_k^{(i)}|x_{k-1}^{(i)},u_{k-1})}{\mathrm{q}(x_k^{(i)}|x_{k-1}^{(i)},u_{k-1})}}{\eta}w_{k-1}^{(i)},$$

with $\eta > 0$ being a normalization factor.

Resampling step. In the following, we discuss the need for the resampling of the particles. It has been shown empirically, that if a large number of weights are very small, then the PF will not work well, if at all. This comes as a consequence of the loss of diversity, i.e. that the small number of high-valued particles may not be able to represent the uncertainty of the process well enough. In other words, this reinforces the bias that only some particles are important, which disregards (to a great extent) the information obtained from other particles, and does not allow the PF to capture changes in the process once the high value particles become less relevant.

This is solved by removing old samples with weights close to zero and replacing them by the particles with higher weights. A typical way to determine if resampling should be done is to first determine a threshold, for instance $N_{tr} = \frac{N}{4}$, and to subsequently check if

$$\frac{1}{\sum_{i=1}^{N}\left(w_k^{(i)}\right)^2} < N_{tr},$$

which if fulfilled indicates the need to perform a resampling step. Alternatively, one could check if

$$\frac{1}{\max_{i=1,..,N}\left(w_k^{(i)}\right)} \leq \frac{1}{0.005}$$

to determine if resampling is to be done. In exact, for the resampling step we consider the set of particles $x_k^{(i)}$, $1 = 1, ..., N$, with their corresponding weights $w_k^{(i)}$. We sample N particles $x_r^{(i)}$ as follows

$$x_r^{(i)} \sim \mathcal{D}\{x_k^{(1)}, ..., x_k^{(N)}\}, \qquad p(x_r^{(i)} = x_k^{(i)}) = w_k^{(i)}.$$

In such a way, it becomes highly likely to sample multiple times the same particle $x_k^{(i)}$ if its weight $w_k^{(i)}$ is significantly higher than the other particles. Moreover, after sampling N new particles, we set $w_k^{(i)} = \frac{1}{N}$, $1 = 1, ..., N$. This means that to each resampled particle we assign the same weight, i.e. influence in determining the estimate.

Estimate mean of the state. In the following, we present one typical estimate of the PF, which is widely used, i.e. the mean-value estimate. From the prediction and update steps (optionally also the resampling one) we now have the posterior density $\hat{\rho}(x_k|u_{0:k-1}, z_{1:k})$, which we now use to estimate the system states, i.e.,

$$\bar{x}_k = \sum_{i=1}^{N} w_k^{(i)} x_k^{(i)},$$

It remains to be mentioned that it is also possible to calculate higher order moments (e.g. variance) from the posterior density, if necessary. However, this is step is more computationally expensive and is therefore often avoided in real-time applications.

The tasks in this problem are as follows. From the information provided above,

2a) explain importance sampling and Monte Carlo integration in your own words,

2b) and summarize the problem of state estimation as well as the derivation of the PF in your own words. Explain the main ideas behind the PF and each step of the derivation of the PF in detail.

Consider now the setup of Problem 1. Provide an answer to following questions:

2c) which terms from Problem 1 correspond to x_k and z_k respectively, and

2d) how the importance density $\mathrm{q}(x_k|x_{k-1}^{(i)}, u_{k-1})$ may be chosen? Provide two possible choices for it.

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Problem 3 In the following we implement the PF algorithm for a state estimation problem. In particular, consider the setup of Problem 1. We aim to implement the PF algorithm in order to estimate the position of the light source. We first present the PF algorithm:

Input : $\rho_{x_0}(\cdot) =: \rho(x_0)$, $\rho_{z_k}(\cdot|x_k) =: \rho(z_k|x_k)$, $\rho_{x_k}(\cdot|x_{k-1}, u_{k-1}) =: \rho(x_k|x_{k-1}, u_{k-1})$, an importance density $\mathrm{q}(\cdot)$, number of samples $N \in \mathbb{N}$, number of time (estimation) steps $T \in \mathbb{N}$, cutoff weight $N_{tr} > 0$, measurement z_k and input u_k (available at time step k), $k \in \{1, ..., T\}$

Output : Estimate $\bar{x}_k$ for every time step $k \in \{1, ..., T\}$

Initialization step:

Sample N particles: $x_0^{(i)} \sim \rho(\cdot)$, $i = 1, \ldots, N$

Set: $w_0^{(i)} = \frac{1}{N}$, $i = 1, \ldots, N$

for $k = 1 : T$ **do**

Prediction and update step:

Sample N particles: $x_k^{(i)} \sim \mathrm{q}(\cdot|x_{k-1}^{(i)}, u_{k-1})$, $i = 1, \ldots, N$

Set: $\hat{\tilde{\omega}}_k^{(i)} = \frac{\rho(z_k|x_k^{(i)})\rho(x_k^{(i)}|x_{k-1}^{(i)}, u_{k-1})}{\mathrm{q}(x_k^{(i)}|x_{k-1}^{(i)}, u_{k-1})} w_{k-1}^{(i)}$, $i = 1, \ldots, N$

$$w_k^{(i)} = \frac{\hat{\tilde{\omega}}_k^{(i)}}{\sum_{j=1}^{N} \hat{\tilde{\omega}}_k^{(j)}}, \; i = 1, \ldots, N$$

Resampling step (optional):

if $\frac{1}{\sum_{i=1}^{N} \left(w_k^{(i)}\right)^2} < N_{tr}$ **then**

Sample N particles $x_r^{(i)}$ from the set of particles $\{x_k^{(1)}, ..., x_k^{(N)}\}$ as follows:

$$x_r^{(i)} \sim \mathcal{D}\{x_k^{(1)}, ..., x_k^{(N)}\} \text{ with } p(x_r^{(i)} = x_k^{(i)}) = w_k^{(i)}$$

(Re)set: $x_k^{(i)} = x_r^{(i)}$, $w_k^{(i)} = \frac{1}{N}$, $i = 1, ..., N$

end

Estimate mean of current state:

Set: $\bar{x}_k = \sum_{i=1}^{N} w_k^{(i)} x_k^{(i)}$

end

Algorithm 1: Particle Filter (PF) Algorithm

The goal is to use the PF to compute an estimate of the position of the light source, i.e. $\xi(t)$ at every time $t \geq 0$ based on the noisy position measurements $\zeta(t)$. Hence, the tasks that need to be addressed are:

3a) to discretize the system dynamics with some $\delta t > 0$ (in seconds), and to choose two parameter pairs $(\sigma_i, \varepsilon_i)$ to model the dynamics as outlined in Problem 1, where $\sigma_i, \varepsilon_i > 0$ for both cases $i = 1, 2$,

3b) to choose an importance density $\mathrm{q}(\cdot)$, and

3c) to implement the PF algorithm in order to estimate the position of the light at every time step for both parameter pairs $(\sigma_i, \varepsilon_i)$, $i = 1, 2$.

To solve this problem, it is required to implement the algorithm for both of the chosen parameter pairs and to perform simulations. The simulation results must be plotted and explained in an

understandable manner. For each pair $(\sigma_i, \varepsilon_i)$, it suffices to simulate the estimation algorithm for two different initial conditions $\xi_0 \in \mathbb{R}^2$ and to plot the (true) position of the light source and estimated light position at every time step. –

Problem 4 In this problem, we address the overall source seeking problem, i.e. estimation of the position of the light using a PF and tracking of the light source by the robot with a suitable control policy. Consider the setup in Problem 1 with the parameters (σ, ε) chosen as in Problem 3 (only one pair is sufficient). We use the PF state estimates in the control policy to steer the robot to the position of the light. In order to achieve this goal, it is firstly necessary to formalize the tracking problem. Specifically, the first task is to determine

4a) the discrete-time dynamics of the robot in the form of

$$\theta_{k+1} = f_2(\theta_k, u_k),$$

4b) and a control policy $u(\theta_k, \xi_k)$ that guarantees the tracking of the light.

Then, based on the setup discussed in Problem 3, the second task is to

4c) implement both, the control policy and the PF, to track the light, and

4d) discuss the effectiveness of the proposed source seeking approach.

For the solution of this problem it is required to implement the algorithms and to carry out simulations. The simulation results for the PF and for the tracking behavior should be plotted in a suitable and understandable manner. It suffices to simulate the estimation and tracking algorithm for two different initial conditions $\xi_0, \theta_0 \in \mathbb{R}^2$ and to graphically plot the true position of the light source, the estimated light position, and the robot position at each time step. –